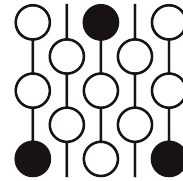


Review of Submission 1C

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. As far as I can tell, the solution is completely correct.

Novelty. This solution is highly reminiscent of a construction introduced by Dan Turetsky in the paper “Coding in the automorphism group of a countable structure” and subsequently used in at least two other papers. It is most similar to a version of Turetsky’s technique used in the proof of Theorem 4.6 of the paper “Scott complexity of countable structures” by Rachel Alvir, Noam Greenberg, Matthew Harrison-Trainor and Dan Turetsky. In particular, the structure $\mathcal{A}$ constructed in the solution below is very similar to the structure constructed by Alvir, Greenberg, Harrison-Trainor and Turetsky, although the purpose it is being used for and the verification that it has the desired properties are somewhat different. I would like to note that this similarity was pointed out to me by another referee, after which I independently verified that the two constructions are indeed highly similar.

It would be interesting to know to what extent the AI system that composed this solution was aware of these papers and, furthermore, to what extent they directly inspired its solution. One piece of evidence which is suggestive, but not definitive, is that some of the notation in the solution below is fairly similar to that used by Turetsky and by Alvir, Greenberg, Harrison-Trainor and Turetsky. For example, in all three, an arbitrary countable infinite dimensional $\mathbb{F}_2$ vector space is needed and the particular one chosen is $([\omega]^{<\omega}, \Delta)$ (i.e. the set of finite sets of natural numbers with the operation of symmetric difference playing the role of vector space addition). This choice is somewhat unusual, though perhaps somewhat less so in computability theory than in some other fields. Furthermore, in both Turetsky’s paper and the solution below, an infinite number of auxiliary sets are needed and in both cases they are referred to as $\{V_n\}_n$. These similarities constitute moderate evidence that the AI system may have been aware of one or both of these papers when devising its solution but fall far short of definitive evidence for this possibility.

If I encountered the solution below as part of a paper I was referring for a journal, I would insist that the authors cite the work of Turetsky and point out the similarity in the constructions, but I would also consider the solution below to be a minor, but novel and interesting variation on Turetsky’s technique.

Exposition. Overall, the quality of the exposition is fairly high. The choice of what statements to separate out into independent lemmas seems especially well-done. There are a few points where things could be done in somewhat simpler ways—for example, in the proof of Proposition 1, there is no need to hard-code u_0 and u_1 into the program and the computation of $\beta(x)$ when $x \in V_n^B$ and $n \in g$ could be made simpler by hard-coding the bottoms of the two chains in V_n^B rather than arbitrary elements of both chains—but these are very minor complaints.

2 Recommendation

Essentially flawless, with one very important caveat. If the AI system was aware of—and especially if it was directly inspired by—Turetsky’s work as discussed above, then its failure to cite or even mention Turetsky (or any of the papers which use his technique) is a serious and very concerning oversight. It seems reasonable likely that this occurred, but not certain.

3 Submission with inline comments

Abstract

We answer the question affirmatively. We build a computable countable structure $\mathcal{A}$ such that every automorphism of every copy of $\mathcal{A}$ is computable from that copy, but for every finite tuple of parameters $\bar{a}$ there is an automorphism π whose graph is not Σ_1^{in} -definable over $\bar{a} \cup \pi(\bar{a})$.

The construction

Let

$$G = [\omega]^{<\omega}$$

be the countable Boolean group of finite subsets of ω , with operation symmetric difference $\triangle$. For $g \in G$ and $n \in \omega$, write

$$\epsilon_n(g) = \begin{cases} 1, & n \in g, \\ 0, & n \notin g. \end{cases}$$

Let $e_n = \{n\}$.

We use a computable countable language L containing:

- unary predicates $U, V_0, V_1, \dots$;
- for each $g \in G$, a binary relation symbol Q_g ;
- for each $n \in \omega$, a binary relation symbol P_n ;
- one unary function symbol S .

The domain of $\mathcal{A}$ is the disjoint union

$$U^{\mathcal{A}} \cup \bigcup_{n \in \omega} V_n^{\mathcal{A}},$$

where

$$U^{\mathcal{A}} = G$$

and

$$V_n^{\mathcal{A}} = V_n^0 \cup V_n^1, \quad V_n^i = \{v_{n,i,m} : m \in \omega\}.$$

The predicates U, V_n name these sorts. The symbols are interpreted as follows:

$$Q_g(u, v) \iff u, v \in U^{\mathcal{A}} \text{ and } v = u \triangle g;$$

$$P_n(u, v_{n,i,m}) \iff u \in U^{\mathcal{A}} \text{ and } i = \epsilon_n(u),$$

and P_n is false outside $U^{\mathcal{A}} \times V_n^{\mathcal{A}}$. Finally,

$$S(u) = u \quad (u \in U^{\mathcal{A}}),$$

and

$$S(v_{n,i,m}) = v_{n,i,m+1}.$$

With any standard computable coding of G and of the triples (n, i, m) , this is a computable presentation of $\mathcal{A}$.

The automorphism group

For each $g \in G$, define a permutation τ_g of $\mathcal{A}$ by

$$\tau_g(u) = g \triangle u \quad (u \in U^{\mathcal{A}}),$$

and

$$\tau_g(v_{n,i,m}) = v_{n,i \oplus \epsilon_n(g),m}.$$

Here $\oplus$ denotes addition modulo 2.

Lemma 1. *For every $g \in G$, $\tau_g \in \text{Aut}(\mathcal{A})$, and every automorphism of $\mathcal{A}$ is of this form. Hence*

$$\text{Aut}(\mathcal{A}) = \{\tau_g : g \in G\} \cong G$$

is countable.

Proof. It is immediate from the definitions that τ_g preserves all sort predicates and preserves S .

If $Q_h(u, v)$ holds, then $v = u \triangle h$, and therefore

$$\tau_g(v) = g \triangle u \triangle h = \tau_g(u) \triangle h,$$

so $Q_h(\tau_g(u), \tau_g(v))$ holds.

Similarly, if $P_n(u, v_{n,i,m})$ holds, then $i = \epsilon_n(u)$. Since

$$\epsilon_n(g \triangle u) = \epsilon_n(g) \oplus \epsilon_n(u),$$

we have

$$P_n(\tau_g(u), \tau_g(v_{n,i,m})).$$

Thus τ_g is an automorphism.

Conversely, let $\alpha \in \text{Aut}(\mathcal{A})$. Since U is named, α maps $U^{\mathcal{A}}$ to itself. Put

$$g = \alpha(\emptyset) \in G.$$

For every $u \in G$, the relation $Q_u(\emptyset, u)$ holds. Applying α , we get

$$Q_u(g, \alpha(u)),$$

so

$$\alpha(u) = g \triangle u.$$

Now fix n and consider V_n . Let $v = v_{n,i,m}$. Choose $u \in G$ with $\epsilon_n(u) = i$. Then $P_n(u, v)$ holds. Applying α , we get

$$P_n(g \triangle u, \alpha(v)).$$

Therefore $\alpha(v)$ lies in the copy $V_n^{i \oplus \epsilon_n(g)}$.

Finally, α preserves S . On each copy V_n^i , the map S is the successor map on a one-way chain of type ω . Any bijection between two such chains preserving successor must send the unique element not in the range of S to the unique element not in the range of S , and hence must preserve the coordinate m . Thus

$$\alpha(v_{n,i,m}) = v_{n,i \oplus \epsilon_n(g),m}.$$

So $\alpha = \tau_g$. □

Every automorphism of every copy is relatively computable

Proposition 1. *For every copy $\mathcal{B} \cong \mathcal{A}$ and every $\beta \in \text{Aut}(\mathcal{B})$, the function β is computable relative to the atomic diagram of $\mathcal{B}$. In particular, $\mathcal{A}$ is computably Aut-countable on the cone with base $\emptyset$.*

Proof. Let $\mathcal{B}$ be any copy of $\mathcal{A}$, and let $\beta \in \text{Aut}(\mathcal{B})$. By the previous lemma, transferred to $\mathcal{B}$, there is a unique $g \in G$ such that β acts on the U -sort as translation by g . Since g is a finite subset of ω , it can be hard-coded into a Turing program computing β .

Choose any $u_0 \in U^{\mathcal{B}}$ and let

$$u_1 = \beta(u_0).$$

The program may hard-code the natural numbers u_0, u_1 .

Given input $x \in U^{\mathcal{B}}$, search through $h \in G$ until finding the unique h such that

$$\mathcal{B} \models Q_h(u_0, x).$$

Then search for the unique y such that

$$\mathcal{B} \models Q_h(u_1, y).$$

Output this y . This computes $\beta(x)$ for $x \in U^{\mathcal{B}}$.

Now suppose $x \in V_n^{\mathcal{B}}$. We can find n by searching the unary predicates $V_0, V_1, \dots$.

If $n \notin g$, then β is the identity on $V_n^{\mathcal{B}}$, so output x .

If $n \in g$, then β swaps the two S -chains inside $V_n^{\mathcal{B}}$. Choose some

$$p_n \in V_n^{\mathcal{B}}$$

and let

$$q_n = \beta(p_n).$$

The program may hard-code the finitely many pairs (p_n, q_n) for $n \in g$.

To compute $\beta(x)$ for $x \in V_n^{\mathcal{B}}$ with $n \in g$, dovetail the following searches:

- search for d with $S^d(p_n) = x$; if found, output $S^d(q_n)$;
- search for d with $S^d(x) = p_n$; if found, search for the unique y with $S^d(y) = q_n$, and output y ;
- search for d with $S^d(q_n) = x$; if found, output $S^d(p_n)$;
- search for d with $S^d(x) = q_n$; if found, search for the unique y with $S^d(y) = p_n$, and output y .

Exactly one of these alternatives eventually applies, because each V_n consists of two one-way S -chains and p_n, q_n are corresponding points in the two chains.

All searches use only the atomic diagram of $\mathcal{B}$. Thus β is computable relative to $\mathcal{B}$. $\square$

A preservation lemma for existential formulas

Fix n . For $d_0, d_1 \in \omega$, define a map

$$\rho_{d_0, d_1}^n : \mathcal{A} \rightarrow \mathcal{A}$$

by fixing every element outside V_n , and by

$$\rho_{d_0, d_1}^n(v_{n, i, m}) = v_{n, i, m + d_i}.$$

Lemma 2. *For every n and every $d_0, d_1 \in \omega$, the map ρ_{d_0, d_1}^n is an embedding of $\mathcal{A}$ into itself. Consequently, if an existential formula with parameters outside V_n holds of a tuple, then it still holds after applying ρ_{d_0, d_1}^n to the non-parameter entries.*

Proof. The map ρ_{d_0, d_1}^n is injective. It clearly preserves the unary predicates. It fixes U , so it preserves every Q_g . It also preserves S , because

$$\rho_{d_0, d_1}^n(S(v_{n, i, m})) = v_{n, i, m + 1 + d_i} = S(\rho_{d_0, d_1}^n(v_{n, i, m})).$$

Finally, $P_n(u, v_{n, i, m})$ depends only on u and on the copy index i , and ρ_{d_0, d_1}^n fixes u and preserves i . For $m \neq n$, P_m is unaffected.

Thus ρ_{d_0, d_1}^n is an L -embedding. Embeddings preserve quantifier-free formulas, and therefore preserve existential formulas. $\square$

Failure of Σ_1^{in} -definability

Lemma 3. *Let $\bar{p}$ be a finite tuple from $\mathcal{A}$ containing no element of V_n . Then the automorphism τ_{e_n} is not Σ_1^{in} -definable over $\bar{p}$.*

Proof. Let

$$r_0 = v_{n,0,0}, \quad r_1 = v_{n,1,0}.$$

Then

$$\tau_{e_n}(r_0) = r_1.$$

Suppose, toward a contradiction, that the graph of τ_{e_n} is Σ_1^{in} -definable over $\bar{p}$. Then there is a countable family of existential formulas

$$(\varphi_i(\bar{p}, x, y))_{i \in \omega}$$

such that

$$\tau_{e_n}(x) = y \iff \mathcal{A} \models \varphi_i(\bar{p}, x, y) \text{ for some } i.$$

Since $\tau_{e_n}(r_0) = r_1$, there is some i with

$$\mathcal{A} \models \varphi_i(\bar{p}, r_0, r_1).$$

Now apply the embedding $\rho_{0,1}^n$. It fixes $\bar{p}$ and r_0 , and sends

$$r_1 \mapsto S(r_1) = v_{n,1,1}.$$

By preservation of existential formulas under embeddings,

$$\mathcal{A} \models \varphi_i(\bar{p}, r_0, S(r_1)).$$

But

$$\tau_{e_n}(r_0) = r_1 \neq S(r_1),$$

so $(r_0, S(r_1))$ is not in the graph of τ_{e_n} . This contradicts the assumed definition of the graph. $\square$

Main theorem

Theorem 1. *There is a countable computably represented structure $\mathcal{A}$ such that:*

1. $\mathcal{A}$ is computably Aut-countable on a cone;
2. for every finite tuple of parameters $\bar{a}$ from $\mathcal{A}$, there is an automorphism $\pi \in \text{Aut}(\mathcal{A})$ such that π is not Σ_1^{in} -definable over $\bar{a} \cup \pi(\bar{a})$.

Proof. The structure $\mathcal{A}$ constructed above is computable and countable. By the proposition, every automorphism of every copy of $\mathcal{A}$ is computable relative to that copy, so $\mathcal{A}$ is computably Aut-countable on the cone with base $\emptyset$.

Now let $\bar{a}$ be any finite tuple from $\mathcal{A}$. Since $\bar{a}$ is finite, it meets only finitely many of the sorts V_n . Choose n such that no entry of $\bar{a}$ lies in V_n .

Let

$$\pi = \tau_{e_n}.$$

Then π toggles the two copies inside V_n and fixes every V_m with $m \neq n$. Since $\bar{a}$ has no entry in V_n , the tuple $\pi(\bar{a})$ also has no entry in V_n . Therefore the finite tuple

$$\bar{p} = \bar{a} \cup \pi(\bar{a})$$

contains no element of V_n .

By the previous lemma, the graph of $\pi = \tau_{e_n}$ is not Σ_1^{in} -definable over $\bar{p}$. Hence π is not Σ_1^{in} -definable over $\bar{a} \cup \pi(\bar{a})$.

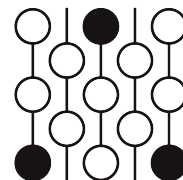
Thus $\mathcal{A}$ has the required properties. □

Review of Submission 1C

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. Problem 1 asks whether there exists a computable structure with certain properties, and this submission correctly constructs an example of such a structure. The structure $\mathcal{A}$ constructed is in a countably infinite language consisting of infinitely many unary and binary relation symbols and one unary function symbol. The proof characterizes the automorphisms of $\mathcal{A}$ by associating each $\alpha \in \mathcal{A}$ to a finite set $g \in [\omega]^{<\omega}$. Finally, it verifies the construction by checking that $\mathcal{A}$ indeed satisfies the required properties. The level of detail in the proof matches the expected level of detail from a journal paper.

Novelty. The proof uses standard techniques from computability theory and computable structure theory. In the opinion of this reviewer, there is no evidence that the solution is following a similar proof from another source. This submission is most similar to Submission B, though Submission C uses an infinite language unlike Submission B's finite relational language. The infinite language makes the construction somewhat cleaner. This submission is less similar to Submission A or the human solution.

Attribution The submission makes no attributions. While this submission does not closely follow or adapt an existing argument, there are somewhat similar arguments in the literature, and those should be cited, such as the following:

- Coding in the Automorphism Group of a Computably Categorical Structure, by Turetsky
- Degrees of Categoricity and Treeable Degrees, by Csima and Rossegger
- Scott Complexity of Countable Structures, by Alvir, Greenberg, Harrison-Trainor, and Turetsky

2 Recommendation

The submission is **publishable with minor revisions**. As stated above, the proof is correct and contains no inaccuracies. The in-text comments below are mostly stylistic disagreements. Additionally, the techniques in the paper are just similar enough to the techniques in other papers

that should be mentioned and cited, even if the argument in this submission does not exactly follow any existing, published argument.

3 Submission with inline comments

Abstract

We answer the question affirmatively. We build a computable countable structure $\mathcal{A}$ such that every automorphism of every copy of $\mathcal{A}$ is computable from that copy, but for every finite tuple of parameters $\bar{a}$ there is an automorphism π whose graph is not Σ_1^{in} -definable over $\bar{a} \cup \pi(\bar{a})$.

The construction

Let

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be the countable Boolean group of finite subsets of ω , with operation symmetric difference $\triangle$. For $g \in G$ and $n \in \omega$, write

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Let $e_n = \{n\}$.

We use a computable countable language L containing:

- unary predicates $U, V_0, V_1, \dots$;
- for each $g \in G$, a binary relation symbol Q_g ;
- for each $n \in \omega$, a binary relation symbol P_n ;
- one unary function symbol S .

Reviewer 2

The structure is in an infinite language, whereas Submission B and the human solution build structures in finite languages. This difference is mostly stylistic but perhaps worth noting. The infinite language does simplify the construction.

The domain of $\mathcal{A}$ is the disjoint union

$$U^{\mathcal{A}} \cup \bigcup_{n \in \omega} V_n^{\mathcal{A}},$$

where

$$U^{\mathcal{A}} = G$$

and

$$V_n^{\mathcal{A}} = V_n^0 \cup V_n^1, \quad V_n^i = \{v_{n,i,m} : m \in \omega\}.$$

The predicates U, V_n name these sorts. The symbols are interpreted as follows:

$$Q_g(u, v) \iff u, v \in U^{\mathcal{A}} \text{ and } v = u \triangle g;$$

$$P_n(u, v_{n,i,m}) \iff u \in U^{\mathcal{A}} \text{ and } i = \epsilon_n(u),$$

and P_n is false outside $U^{\mathcal{A}} \times V_n^{\mathcal{A}}$. Finally,

$$S(u) = u \quad (u \in U^{\mathcal{A}}),$$

and

$$S(v_{n,i,m}) = v_{n,i,m+1}.$$

With any standard computable coding of G and of the triples (n, i, m) , this is a computable presentation of $\mathcal{A}$.

The automorphism group

For each $g \in G$, define a permutation τ_g of $\mathcal{A}$ by

$$\tau_g(u) = g \triangle u \quad (u \in U^{\mathcal{A}}),$$

and

$$\tau_g(v_{n,i,m}) = v_{n,i \oplus \epsilon_n(g),m}.$$

Here $\oplus$ denotes addition modulo 2.

Lemma 1. *For every $g \in G$, $\tau_g \in \text{Aut}(\mathcal{A})$, and every automorphism of $\mathcal{A}$ is of this form. Hence*

$$\text{Aut}(\mathcal{A}) = \{\tau_g : g \in G\} \cong G$$

is countable.

Proof. It is immediate from the definitions that τ_g preserves all sort predicates and preserves S .

Reviewer 2

Indeed, it is appropriate at this level to not need to check with great detail that τ_g preserves those relations.

If $Q_h(u, v)$ holds, then $v = u \triangle h$, and therefore

$$\tau_g(v) = g \triangle u \triangle h = \tau_g(u) \triangle h,$$

so $Q_h(\tau_g(u), \tau_g(v))$ holds.

Similarly, if $P_n(u, v_{n,i,m})$ holds, then $i = \epsilon_n(u)$. Since

$$\epsilon_n(g \triangle u) = \epsilon_n(g) \oplus \epsilon_n(u),$$

we have

$$P_n(\tau_g(u), \tau_g(v_{n,i,m})).$$

Thus τ_g is an automorphism.

Conversely, let $\alpha \in \text{Aut}(\mathcal{A})$. Since U is named, α maps $U^{\mathcal{A}}$ to itself. Put

$$g = \alpha(\emptyset) \in G.$$

For every $u \in G$, the relation $Q_u(\emptyset, u)$ holds. Applying α , we get

$$Q_u(g, \alpha(u)),$$

so

$$\alpha(u) = g \triangle u.$$

Now fix n and consider V_n . Let $v = v_{n,i,m}$. Choose $u \in G$ with $\epsilon_n(u) = i$. Then $P_n(u, v)$ holds. Applying α , we get

$$P_n(g \triangle u, \alpha(v)).$$

Therefore $\alpha(v)$ lies in the copy $V_n^{i \oplus \epsilon_n(g)}$.

Finally, α preserves S . On each copy V_n^i , the map S is the successor map on a one-way chain of type ω . Any bijection between two such chains preserving successor must send the unique element not in the range of S to the unique element not in the range of S , and hence must preserve the coordinate m . Thus

$$\alpha(v_{n,i,m}) = v_{n,i \oplus \epsilon_n(g),m}.$$

So $\alpha = \tau_g$. □

Reviewer 2

Compare to Lemmas 3.5 and 3.6 of Submission B.

Every automorphism of every copy is relatively computable

Proposition 1. *For every copy $\mathcal{B} \cong \mathcal{A}$ and every $\beta \in \text{Aut}(\mathcal{B})$, the function β is computable relative to the atomic diagram of $\mathcal{B}$. In particular, $\mathcal{A}$ is computably Aut -countable on the cone with base $\emptyset$.*

Proof. Let $\mathcal{B}$ be any copy of $\mathcal{A}$, and let $\beta \in \text{Aut}(\mathcal{B})$. By the previous lemma, transferred to $\mathcal{B}$, there is a unique $g \in G$ such that β acts on the U -sort as translation by g . Since g is a finite subset of ω , it can be hard-coded into a Turing program computing β .

Choose any $u_0 \in U^{\mathcal{B}}$ and let

$$u_1 = \beta(u_0).$$

The program may hard-code the natural numbers u_0, u_1 .

Given input $x \in U^{\mathcal{B}}$, search through $h \in G$ until finding the unique h such that

$$\mathcal{B} \models Q_h(u_0, x).$$

Then search for the unique y such that

$$\mathcal{B} \models Q_h(u_1, y).$$

Output this y . This computes $\beta(x)$ for $x \in U^{\mathcal{B}}$.

Now suppose $x \in V_n^{\mathcal{B}}$. We can find n by searching the unary predicates $V_0, V_1, \dots$.

If $n \notin g$, then β is the identity on $V_n^{\mathcal{B}}$, so output x .

If $n \in g$, then β swaps the two S -chains inside $V_n^{\mathcal{B}}$. Choose some

$$p_n \in V_n^{\mathcal{B}}$$

and let

$$q_n = \beta(p_n).$$

The program may hard-code the finitely many pairs (p_n, q_n) for $n \in g$.

To compute $\beta(x)$ for $x \in V_n^{\mathcal{B}}$ with $n \in g$, dovetail the following searches:

- search for d with $S^d(p_n) = x$; if found, output $S^d(q_n)$;
- search for d with $S^d(x) = p_n$; if found, search for the unique y with $S^d(y) = q_n$, and output y ;
- search for d with $S^d(q_n) = x$; if found, output $S^d(p_n)$;
- search for d with $S^d(x) = q_n$; if found, search for the unique y with $S^d(y) = p_n$, and output y .

Exactly one of these alternatives eventually applies, because each V_n consists of two one-way S -chains and p_n, q_n are corresponding points in the two chains.

All searches use only the atomic diagram of $\mathcal{B}$. Thus β is computable relative to $\mathcal{B}$.

Reviewer 2

There are some unnecessary details in this proof but no inaccuracies.

□

A preservation lemma for existential formulas

Fix n . For $d_0, d_1 \in \omega$, define a map

$$\rho_{d_0, d_1}^n : \mathcal{A} \rightarrow \mathcal{A}$$

by fixing every element outside V_n , and by

$$\rho_{d_0, d_1}^n(v_{n, i, m}) = v_{n, i, m + d_i}.$$

Lemma 2. *For every n and every $d_0, d_1 \in \omega$, the map ρ_{d_0, d_1}^n is an embedding of $\mathcal{A}$ into itself. Consequently, if an existential formula with parameters outside V_n holds of a tuple, then it still holds after applying ρ_{d_0, d_1}^n to the non-parameter entries.*

Proof. The map ρ_{d_0, d_1}^n is injective. It clearly preserves the unary predicates. It fixes U , so it preserves every Q_g . It also preserves S , because

$$\rho_{d_0, d_1}^n(S(v_{n, i, m})) = v_{n, i, m + 1 + d_i} = S(\rho_{d_0, d_1}^n(v_{n, i, m})).$$

Finally, $P_n(u, v_{n, i, m})$ depends only on u and on the copy index i , and ρ_{d_0, d_1}^n fixes u and preserves i . For $m \neq n$, P_m is unaffected.

Thus ρ_{d_0, d_1}^n is an L -embedding. Embeddings preserve quantifier-free formulas, and therefore preserve existential formulas. □

Failure of Σ_1^{in} -definability

Lemma 3. *Let $\bar{p}$ be a finite tuple from $\mathcal{A}$ containing no element of V_n . Then the automorphism τ_{e_n} is not Σ_1^{in} -definable over $\bar{p}$.*

Proof. Let

$$r_0 = v_{n,0,0}, \quad r_1 = v_{n,1,0}.$$

Then

$$\tau_{e_n}(r_0) = r_1.$$

Suppose, toward a contradiction, that the graph of τ_{e_n} is Σ_1^{in} -definable over $\bar{p}$. Then there is a countable family of existential formulas

$$(\varphi_i(\bar{p}, x, y))_{i \in \omega}$$

such that

$$\tau_{e_n}(x) = y \iff \mathcal{A} \models \varphi_i(\bar{p}, x, y) \text{ for some } i.$$

Since $\tau_{e_n}(r_0) = r_1$, there is some i with

$$\mathcal{A} \models \varphi_i(\bar{p}, r_0, r_1).$$

Now apply the embedding $\rho_{0,1}^n$. It fixes $\bar{p}$ and r_0 , and sends

$$r_1 \mapsto S(r_1) = v_{n,1,1}.$$

By preservation of existential formulas under embeddings,

$$\mathcal{A} \models \varphi_i(\bar{p}, r_0, S(r_1)).$$

But

$$\tau_{e_n}(r_0) = r_1 \neq S(r_1),$$

so $(r_0, S(r_1))$ is not in the graph of τ_{e_n} . This contradicts the assumed definition of the graph. $\square$

Main theorem

Theorem 1. *There is a countable computably represented structure $\mathcal{A}$ such that:*

1. $\mathcal{A}$ is computably Aut-countable on a cone;
2. for every finite tuple of parameters $\bar{a}$ from $\mathcal{A}$, there is an automorphism $\pi \in \text{Aut}(\mathcal{A})$ such that π is not Σ_1^{in} -definable over $\bar{a} \cup \pi(\bar{a})$.

Proof. The structure $\mathcal{A}$ constructed above is computable and countable. By the proposition, every automorphism of every copy of $\mathcal{A}$ is computable relative to that copy, so $\mathcal{A}$ is computably Aut-countable on the cone with base $\emptyset$.

Now let $\bar{a}$ be any finite tuple from $\mathcal{A}$. Since $\bar{a}$ is finite, it meets only finitely many of the sorts V_n . Choose n such that no entry of $\bar{a}$ lies in V_n .

Let

$$\pi = \tau_{e_n}.$$

Then π toggles the two copies inside V_n and fixes every V_m with $m \neq n$. Since $\bar{a}$ has no entry in V_n , the tuple $\pi(\bar{a})$ also has no entry in V_n . Therefore the finite tuple

$$\bar{p} = \bar{a} \cup \pi(\bar{a})$$

contains no element of V_n .

By the previous lemma, the graph of $\pi = \tau_{e_n}$ is not Σ_1^{in} -definable over $\bar{p}$. Hence π is not Σ_1^{in} -definable over $\bar{a} \cup \pi(\bar{a})$.

Thus $\mathcal{A}$ has the required properties. □